

The Signaling Gap

State-Owned Media and the Strategic Management of
Sentiment during the Russian Invasion of Ukraine

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The Puzzle

Constrained Communication

When exit is too costly and direct voice risks retaliation, actors seek indirect channels to communicate dissatisfaction.

- **Hirschman (1970):** Exit, Voice, and Loyalty — when exit and voice are both foreclosed, actors search for alternative channels
- **Kassing (1997):** Organizational dissent — subordinates route disagreement to external audiences who have leverage over the principal
- Logic is general: any hierarchical relationship where a subordinate depends on a patron but must also manage other audiences

In international politics, the stakes are highest.

When Formal Channels Become Liabilities

States that depend on a powerful patron for their security **do not always have the luxury of saying what they think.**

- Formal channels (UN votes, official statements) risk retaliation from the patron
- But **silence risks being seen as endorsement** when the patron draws international condemnation

Question: What channels do constrained states use to signal distance from a patron's position, and when are they credible?

A Gap Between Scholars and Practitioners

Political Science

State media studied as a **domestic** tool:

- Agenda setting (McCombs 2002)
- Spin dictators (Guriev & Treisman 2022)
- Autocratic media management (Egorov et al. 2009)

Orientation: how governments shape what *their own publics* think

Intelligence Practice

State media monitored as **international signal**:

- BBC Monitoring (est. 1939)
- FBIS / Kremlinology (Cold War)
- OSINT formalized (Manor 2019; USIC 2024)

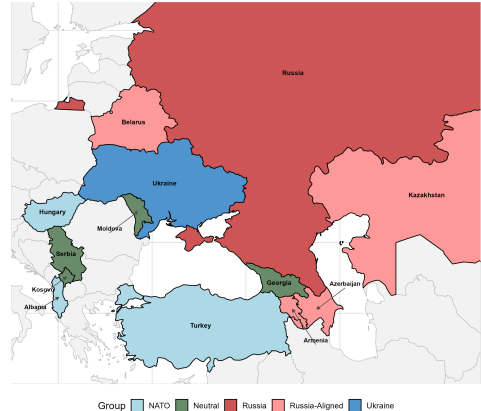
Orientation: reading state media as a window into *regime preferences*

Same outlet does both. Tools exist — but haven't been assembled.

The Russian invasion of Ukraine forced Russia's security partners to navigate this bind.

- Russia-aligned states had reasons to oppose the invasion
- Violated sovereignty norms that protect small states
- Threatened substantial Western trade ties
- But formal condemnation of a patron willing to use force → not viable

Study Region: Russia's Near Abroad



The formal record: UN General Assembly vote condemning the invasion, March 2, 2022

	State	Vote
Russia-Aligned	Belarus	Against
	Armenia	Abstained
	Kazakhstan	Abstained
	Azerbaijan	Absent
Neutral	Georgia, Moldova, Serbia	In favor
	Kosovo	<i>Not a UN member</i>

Why this case: a sharp discontinuity, a well-defined alliance structure, and competing predictions the data can adjudicate.

Argument: State-owned media provides an institutionally mediated signaling channel through which constrained states signal distance from their patron's position to external audiences.

Roadmap: Theory → Data → Design → Results → Implications

This Paper

Preview of findings:

1. The invasion produced a significant negative shift in media sentiment toward Russia across the region
2. State-owned media shifted *less* negatively than independent media → strategic management
3. State media in Russia-*aligned* states shifted *more* negatively than neutral states — **opposite of domestic suppression prediction**

These states refused to formally condemn the invasion while allowing their state media to turn sharply critical.

Theory

Signaling Under Constraint

Standard signaling theory: credible signals require costs (Fearon 1997)

- Audience costs: leaders stake political capital on public positions
- Sunk costs: troop deployments, sanctions, arms transfers
- Common feature: **visible, attributable, formal** channels

But alliances constrain:

- Alliance membership trades autonomy for security (Morrow 1991)
- Public dissent risks abandonment (Snyder 1997)
- Formal channels lock precisely when dissent is most needed (Weitsman 2004)

→ **What do constrained states do when formal channels are unavailable?**

The Signaling Mechanism

State media signals work through two properties:

1. Readable

- Common knowledge of state control → deviation from baseline is informative (Gehlbach et al. 2014)
- Foreign analysts read shifts as regime choice, not journalistic noise
- Why not just loosen independent media? Because independent shifts aren't readable as a *regime choice* — only state media carries the attribution

The Signaling Mechanism

State media signals work through two properties:

2. Costly → Belief-Shifting

- Allowing negative coverage of a security patron risks provoking retaliation
- A state with no reason to signal distance would not accept this risk → bluffing is irrational
- The signal shifts the receiver's beliefs about alignment — it doesn't require revealing "true" preferences, only that foreign analysts assign higher probability to non-alignment after observing the shift

Costs, Benefits, and Sustainability

The strategic calculus:

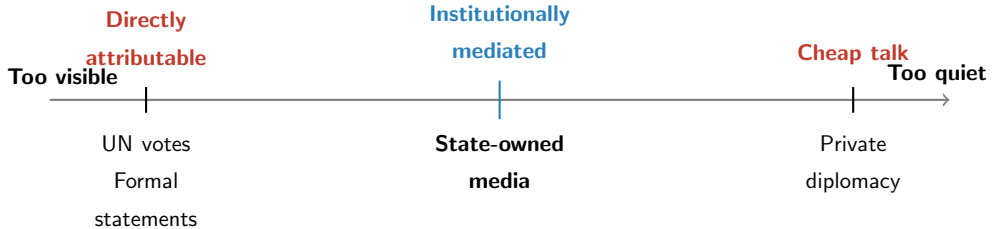
- **Cost of signaling:** Risk of patron retaliation — Russia has demonstrated its willingness to use force in the region (Georgia 2008, Ukraine 2014)
- **Benefit of signaling:** Shifting Western beliefs about alignment → sanctions treatment, diplomacy, trade preservation
- **Cost of not signaling:** Being categorized alongside an internationally condemned patron

Costs, Benefits, and Sustainability

Sustainability (scope condition — even a one-shot signal is valuable):

- **Formal-break threshold:** State media stays below the line that triggers patron punishment. Institutional separation between government and editorial function provides deniability — coverage can be disavowed as editorial judgment rather than state policy (Entman 2004)
- **Mutual dependence:** Russia values the CSTO for strategic reasons (Allison 2008). Punishing a client state for media-level dissent risks pushing that state toward the West — precisely the outcome Russia's alliance framework is designed to prevent

Why State Media Rather Than Alternatives?



- Public enough to be observed by foreign governments that systematically monitor it
- Below the formal-break threshold that triggers patron retaliation
- Readable as a regime choice because external observers know state media reflects regime preferences

Hypotheses

H1: An internationally condemned action by the patron produces a significant decline in media sentiment toward the patron across both state-owned and independent outlets.

H2: State-owned media will report negative sentiment toward the patron after the shock to a lesser extent than independent media.

H3: State-owned media in states formally allied to the patron will respond more negatively to the shock than state-owned media in neutral states.

Hypotheses

H1: An internationally condemned action by the patron produces a significant decline in media sentiment toward the patron across both state-owned and independent outlets.

Without a baseline negative shock, the divergence in H2 and H3 would be uninterpretable.

H2: State-owned media will report negative sentiment toward the patron after the shock to a lesser extent than independent media.

H3: State-owned media in states formally allied to the patron will respond more negatively to the shock than state-owned media in neutral states.

Hypotheses

H1: An internationally condemned action by the patron produces a significant decline in media sentiment toward the patron across both state-owned and independent outlets.

H2: State-owned media will report negative sentiment toward the patron after the shock to a lesser extent than independent media.

A differential response establishes that state media operates under political constraints — this is what makes coverage readable as a regime choice for H3.

H3: State-owned media in states formally allied to the patron will respond more negatively to the shock than state-owned media in neutral states.

Hypotheses

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Domestic suppression and signaling generate opposite predictions — this is the critical test.

Competing Predictions for H3

Domestic Suppression

- Patron-aligned states **suppress** negative coverage to protect the alliance
- State media *less* negative than in neutral states

Signaling

- Patron-aligned states use state media to **signal distance** because formal channels are foreclosed
- State media *more* negative than in neutral states

Opposite predictions provide leverage for distinguishing between mechanisms.

A third account — pretext exposure: Russia framed the invasion as protecting ethnic Russians abroad, giving states with large ethnic Russian minorities their own reason to oppose it. If coverage tracks ethnic Russian share rather than alignment, that vulnerability, not the alliance constraint, should explain the pattern. We test this with an alternative partition.

Data & Measurement

Sample Overview

Corpus: HQMARC (Moratz et al. 2025)

Coverage: Aug 2021 – Aug 2022

Languages: Neural machine translation

Sources: 61 outlets, 11 countries

Articles: ~174,000 Russia-related (excl. Ukraine)

Unit: Article, timestamped to the minute

Russia-Aligned	Armenia, Azerbaijan, Belarus, Kazakhstan
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Neutral	Georgia, Moldova, Kosovo, Serbia
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NATO	Albania, Hungary, Turkey
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State-owned vs. independent

- State-owned: owned and operated by the state or by close affiliates of the executive (family, cabinet ministers)
- Traced through ownership reports, press freedom assessments, and regional experts
- 17 state-owned, 44 independent; at least one state-owned outlet per country

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Neutral	Georgia, Moldova, Kosovo, Serbia
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Comparison logic

- **Neutral states are the H3 baseline:**
formal channels open, so no signaling need; a mild incentive not to antagonize Russia should keep coverage flat
- **NATO members are held out of H3:** an alliance-based reason to oppose the invasion that moves coverage regardless of signaling

Measurement: GPT-4o Classification

Coded at the article level (title + first two sentences), zero-shot:

1. **Topic:** Political, Economic, Military, Cultural, Other
2. **Sentiment toward Russia** (the state, military, government, Putin): -1 , 0 , $+1$

Why a prompted LLM? The target is the article's *orientation toward Russia*, not the tone of the events it describes. An article can report the war neutrally, criticize Russia, or frame Russia's actions sympathetically — off-the-shelf sentiment classifiers cannot separate these.

Measurement: GPT-4o Classification

Validation: 100 articles, adversarially double-coded by the coauthors and debated to consensus

Dimension	F1
Sentiment	0.92
Topic	0.80

Errors work against us: every classifier–human disagreement is between adjacent categories (positive/neutral or negative/neutral), never positive/negative. Measurement error compresses sentiment toward neutral, attenuating the shifts we estimate.

What Do These Articles Look Like?

Positive Sentiment Toward Russia

Country: Belarus

Outlet: *eng.belta.by* (State-owned)

Date: 2022-04-12

“Lukashenko: Belarus has something to contribute to Russia's might.”

“Belarus and Russia will surely endure in the current circumstances. Belarus President Aleksandr Lukashenko expressed confidence about it during negotiations with President of Russia Vladimir Putin at the Vostochny Cosmodrome in Russia's Amur Oblast on 12 April.”

What Do These Articles Look Like?

Neutral Sentiment Toward Russia

Country: Armenia

Outlet: *golosarmenii.am* (Independent)

Date: 2022-04-01

“Armenia abstained in the UNHRC vote on unilateral sanctions.”

“Twenty-seven of the 47 countries, including Russia, China and India, voted in favour of the relevant resolution adopted at the meeting on Thursday, while 14 States, including the United States, the European Union, the United Kingdom and Ukraine, objected. Six delegations abstained in the vote, including Armenia.”

What Do These Articles Look Like?

Negative Sentiment Toward Russia

Country: Serbia

Outlet: *danas.rs* (Independent)

Date: 2022-05-26

“De-Ukrainization in Mariupol: Russians installed mobile TV propaganda, children abolished holidays.”

“The Russian forces that occupied Mariupol have decided to cancel the summer vacation so that the students can master the Russian program, said today the mayor’s adviser Petro Andryushchenko. Children will have classes in Russian language and literature, Russian history and mathematics all summer.”

One Outlet, Two Months: Public Radio of Armenia

January 2022 coded **positive**

“Opening of regional communications to give Armenia rail access to Russia, Europe, Iran, China”

The prime minister's infrastructure plans; Russia is one economic partner among several.

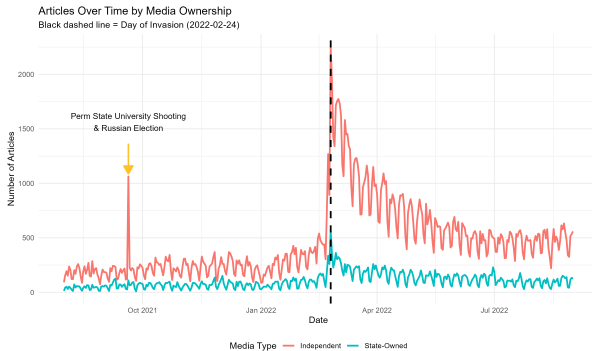
March 2022 coded **negative**

“Russia walks out of Council of Europe”

Wire-style reporting of Russia's rupture with European institutions Armenia belongs to.

Neither article editorializes. What changed is which Russia coverage a state outlet selected and foregrounded — readable as a choice. Over the same weeks, Armenia abstained on the UN resolution condemning the invasion.

Article Volume Over Time



The spike at the invasion is visible in both media types. Independent media produces substantially more Russia-related coverage throughout — which is why the analysis standardizes sentiment within each source.

Research Design

Identification Strategy

We compare how outlets wrote about Russia in the days just before February 24 with the days just after.

- **Regression discontinuity in time.** The buildup was gradual; the full-scale invasion was discrete, and its timing was unknown to the outlets
- **The estimand bundles** the invasion and what it set in motion — sanctions, ruptures, war coverage. These are treatment, not confounders
- **Running variable:** publication date-time, dense to the minute. Missing timestamps are drawn from the source's own publishing rhythm

Identification Strategy

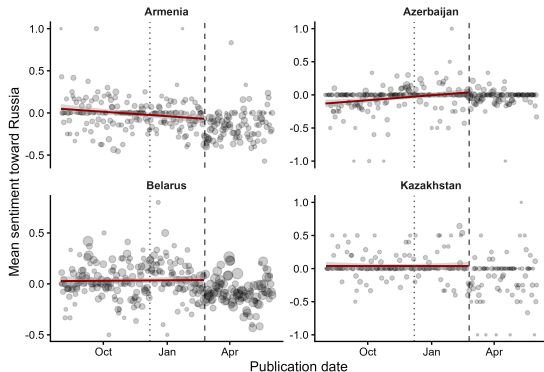
Three levels of analysis:

1. **H1 — RDD:** effect of the invasion on sentiment, all media
2. **H2 — Diff-in-Disc:** State-Owned $\times$ Treatment, full sample
3. **H3 — Diff-in-Disc:** Russia-Aligned $\times$ Treatment, state media only

Key design choices:

- MSE-optimal bandwidth via `rdrobust`; local linear, triangular kernel
- Sentiment z-scored within source, plus topic fixed effects
- Source-level clustering; wild cluster restricted bootstrap where clusters are few

Continuity: Pre-Trends and the December Warning



Russia-aligned state media, pre-invasion:

- Trends are small and mixed in sign; the only two slopes distinguishable from zero point in opposite directions
- **Placebo cutoff at December 7** (US public warning): pooled estimate 0.108, $p = 0.225$
- Only Belarus is individually significant, and it is *positive* (0.320, $p = 0.025$)

December 7 sits about three bandwidths from the cutoff. No anticipatory decline in sentiment in the group where it would matter most. [► Placebo Tables](#)

Specification: Base RDD (H1)

$$Y_p^z = \beta_0 + \beta_1 D_p + \tau T_p + \beta_3 D_p T_p \\ + \beta_4 G_p + \beta_5 T_p G_p + \beta_6 D_p G_p + \beta_7 D_p T_p G_p + \gamma_{\text{topic}} + \varepsilon_p$$

Y_p^z Sentiment of article p , z-scored within source

D_p Running variable (publication date-time, relative to Feb 24)

T_p Post-invasion indicator

τ **Coefficient of interest:** effect of the invasion

G_p Group indicator (introduced in H2/H3)

Estimated with `rdrobust`; MSE-optimal bandwidth; SEs clustered at the source level.

Specification: Diff-in-Disc (H2, H3)

$$Y_p^z = \beta_0 + \beta_1 D_p + \tau T_p + \beta_3 D_p T_p \\ + \beta_4 G_p + \beta_5 T_p G_p + \beta_6 D_p G_p + \beta_7 D_p T_p G_p + \gamma_{\text{topic}} + \varepsilon_p$$

G_p Group indicator: State-Owned (H2); Russia-Aligned, state media only (H3)

β_5 **Coefficient of interest:** differential shift at the cutoff

Parallel discontinuities: absent the group-specific mechanism, both groups jump by the same amount. Levels (β_4) and differential trends (β_6) are absorbed, so anticipatory drift cannot drive a differential jump — and any pre-invasion adjustment by aligned state media works *against* H3.

OLS within the MSE-optimal bandwidth; WCR bootstrap with Webb weights for β_5 ($B = 99,999$, null imposed).

What the Diff-in-Disc Identifies

The within-group drops are causal; the gap between groups is a robust comparison, not a causal effect of alliance membership.

- **Within-group RDD:** causal under continuity
- **Cross-group RDID:** a differential *conditional on* group membership, not *caused by* it (Bansak et al. 2022)
- **Why not causal?** That needs a *discontinuity of differences* — the same units before and after a reform, differenced within unit (Grembi et al. 2016; Tramontin & Shinoki 2024). We have different groups at one cutoff: a *difference of discontinuities*. The threat is a factor correlated with group membership that produces its own jump at the invasion

What the Diff-in-Disc Identifies

Three reasons to read the differential as the mechanism, not a confounder:

1. **Sensitivity bound.** After matching on date and topic, a confounder would need to be $1.5\times$ as strongly associated with sentiment as the invasion itself to nullify the differential
2. **Opposite-sign predictions.** A confounder would have to push the differential in the direction that contradicts the domestic account
3. **Alternative partition.** Splitting states by ethnic Russian share instead of alignment yields no differential

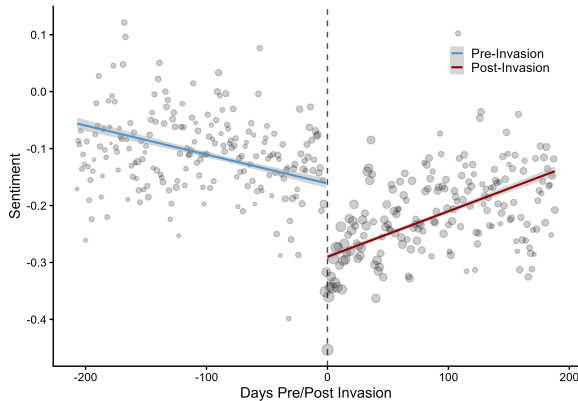
► Total vs. Direct Effect

► Sensitivity

► Alternative Partition

Results

H1: The Invasion Produced a Negative Sentiment Shock



► Full Specifications

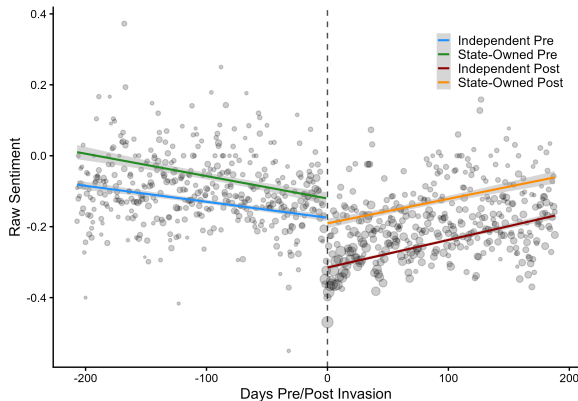
► Full-Sample Figure

Invasion: -0.184 SD

$p < 0.001$. Z-scored sentiment, topic FEs, non-military articles, MSE-optimal bandwidth, source-clustered SEs (Table 1, Model 6)

- Raw $-1/0/+1$ scale: daily mean sentiment drops from about -0.15 to -0.30 at the cutoff
- Stable across fixed-effects specifications; the non-military estimate is the conservative one

H2: State Media Managed Coverage More Carefully



State-Owned $\times$ Treat: +0.222 SD

$p < 0.05$, WCR bootstrap, 43 clusters.

Non-NATO, non-military, matched on date and topic (Table 2, Model 6)

- Separate RDDs: independent -0.214 SD ($p < 0.001$); state-owned -0.109 SD ($p < 0.1$)
- Independent media shifts about twice as far; the gap holds within every topic

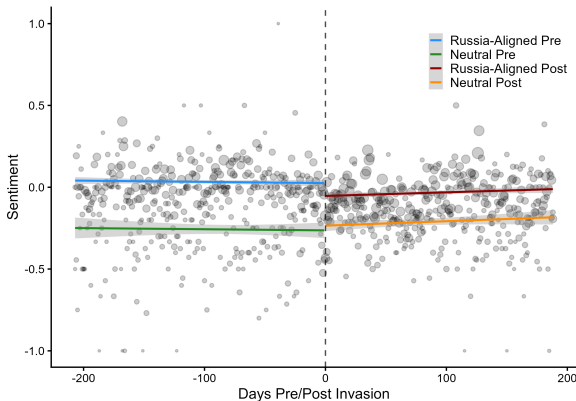
► Full Specifications

► By Topic

► Floor Effect

► Serbia Pair

H3: Russia-Aligned State Media Shifted More Negatively



Russia-Aligned $\times$ Treat: -0.38 SD

$p < 0.05$, WCR bootstrap, 13 clusters. State media only, non-NATO, non-military, matched on date and topic (Table 4, Model 5)

- Separate RDDs: aligned -0.220 SD ($p < 0.05$); neutral 0.055 (n.s.)
- Raw scale: aligned falls from about 0 to -0.08 ; neutral is flat — the opposite of what domestic suppression predicts

► Full Specifications

► Alternative Partition

► Leave-One-Out

Putting It Together

Hypothesis	Prediction	Result
H1: Baseline shock	Negative shift	Confirmed (−0.18 to −0.28 SD)
H2: State vs. Independent	State less negative	Confirmed (+0.22 SD)
H3: Russia-aligned vs. Neutral	Aligned <i>more</i> negative	Confirmed (−0.33 to −0.38 SD)

- H3 is the *opposite* of what domestic audience management predicts
- Partitioning by ethnic Russian share instead of alignment produces no differential — the pattern tracks the alliance constraint, not pretext exposure ▶ Alternative Partition
- The states whose formal channels were most constrained produced the sharpest media signal

Robustness

Check	Result	
Placebo cutoff, Dec 7, 2021 (aligned state media, pooled)	0.108, $p = 0.225$; no pre-invasion jump	► Tables
Matched sensitivity (Cinelli & Hazlett 2020)	A confounder explaining 56% of the interaction's residual variance would still need 1.5× the invasion's own association with sentiment	► Plot
Alternative partition (ethnic Russian share)	Interaction -0.055 to 0.015 , never significant	► Table
Leave-one-out by country and by source	Stable; dropping Belarus attenuates H3, as expected	► Table
Country-level clustering (11 / 8 clusters)	H2 $p = 0.021$; H3 $p = 0.032$	
Bandwidth, 0.5× to 1.5× optimal	Stable	► Figures
Total effect (no topic adjustment)	H2: 0.208 vs. 0.222; H3: -0.37 vs. -0.33	► Detail
Forcing variable (date only; drop imputed times)	Every estimate keeps its sign; H2 and H3 lose some precision	► Tables

Discussion

What the Findings Mean

States whose formal channels are constrained by dependence on a powerful patron produce systematically different media signals than states whose channels remain open.

- State media sentiment is not noise — it is a legible instrument of international communication
- The signal shifts receivers' beliefs about alignment while staying below the formal-break threshold; mutual dependence keeps the channel open
- Western trade with Russia-aligned states held steady through 2022 [▶ Trade Data](#)

Cohen (1987) was right. The channel Soviet-aligned states used during the Cold War remains operative — and is now empirically identifiable.

Contributions

1. **Bridging the theory-practice gap.** Signals practitioners have monitored for 80+ years are theoretically structured and empirically identifiable — they follow from the logic of alliance constraints
2. **Institutionally mediated signaling under alliance constraints.** When formal channels are foreclosed, states turn to channels below the formal-break threshold — a logic that extends to any patron-client hierarchy
3. **Methodological.** Prompted LLM classification at scale (174K articles, $F1 = 0.92$) that targets orientation toward a specific actor, paired with RD in time; and a matched sensitivity procedure for cross-group discontinuity comparisons — coarsened exact matching, then Cinelli & Hazlett (2020) bounds on the differential ► Sensitivity

Limitations

- **A single crisis.** The invasion offers unusual analytical clarity; whether the same logic operates under other alliance structures is the next test
- **Sender, not receiver.** We identify the signal sent, not whether it was detected, interpreted, or acted upon — though the monitoring infrastructure is well established
- **Strategic direction or factional pressure?** The data cannot distinguish them. Either process produces a costly, readable signal of distance from the patron, and the design identifies the shift regardless

Future Work

- **The USAID cuts as a shock to media sentiment toward the United States.** Did outlets that had received USAID funding shift their coverage of the United States differently from outlets that had not? (*Dissertation paper 3*)
- **Group dynamics in elections (RCVD).** Extending the framework to shifts across precincts — how they aggregate to wards and districts, and how to measure those changes across multiple elections (*Dissertation paper 2*)
- **Science fiction and conflict in space.** How do a country's depictions of space shape public support for escalation in a space conflict? A six-country survey experiment paired with text analysis of each country's science fiction

Thank You

Appendix

B. Data and Measurement

Measurement: GPT-4o Classification

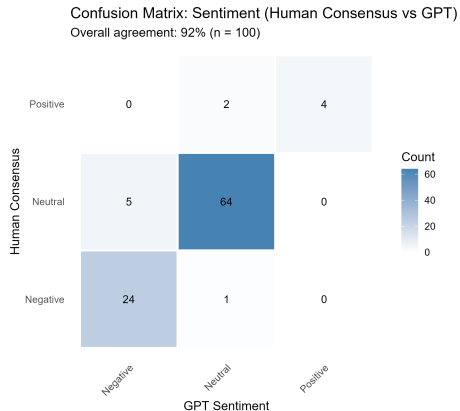
Three dimensions coded at article level (title + first two sentences):

1. **Topic:** Political, Economic, Military, Cultural, Other
2. **Russia involvement:** Binary (main actor or not); supplementary analyses only
3. **Sentiment toward Russia:** Ternary (-1 , 0 , $+1$)

Validation: 100 adversarially double-coded articles, debated to consensus

Dimension	GPT F1 Score
Sentiment	0.92
Russia involvement	0.83
Topic	0.80

GPT Classifier: Sentiment Confusion Matrix



92% overall agreement ($n = 100$). Adversarial double-coding by coauthors with consensus resolution. All disagreement is between adjacent categories, so misclassification biases toward null.

Same War, Same Country, Opposite Orientation

Independent — N1, February 25 coded **negative**

“Canak: Putin started the attack on the same day that Hitler presented his program”

A Serbian politician's charge that Putin launched the invasion on the anniversary of Hitler's Nazi program.

State-owned — RTS, February 26 coded **neutral**

“Bocan-Harchenko: Russia understands Serbia's position, does not require anything from Belgrade”

A report built around the Russian ambassador's assurances to Belgrade.

The medium is held constant — both Serbian outlets covering the invasion in real time — so the gap in orientation tracks ownership rather than the event. Translated from Serbian.

Ethnic Russian Population × Security Alignment

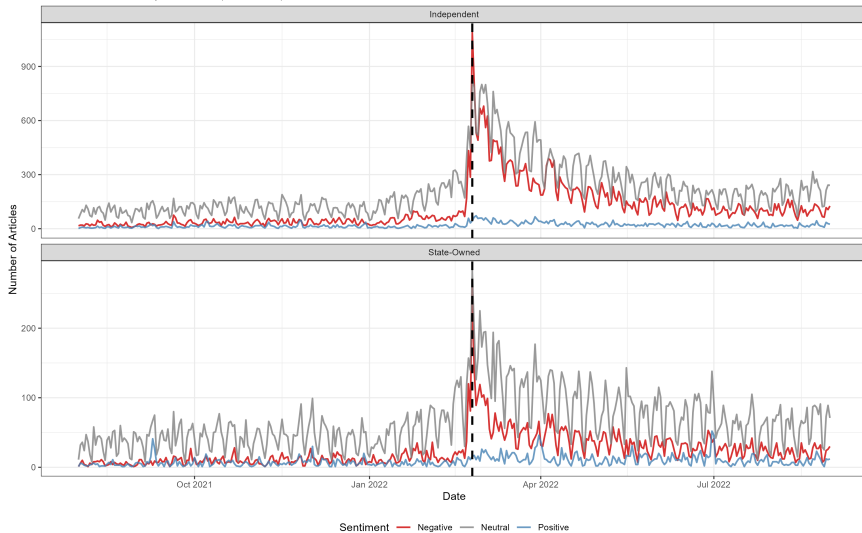
	Neutral	Russia-Aligned
Low	Kosovo, Serbia	Azerbaijan, Armenia
High	Moldova, Georgia	Belarus, Kazakhstan

The two dimensions cross-cut cleanly, so a result that loads on alignment but not on ethnic Russian share isolates the alliance constraint rather than pretext exposure or cultural proximity. [► Results](#)

Daily Sentiment Counts by Media Type

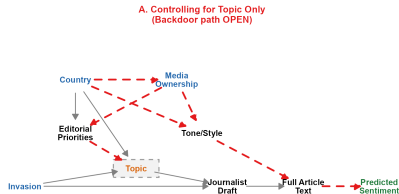
Daily Sentiment Article Counts by Media Type

Black dashed line = Day of Invasion (2022-02-24)



C. Research Design

The Identification Problem

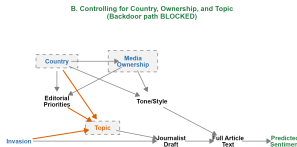


The invasion changes both *how* outlets write about Russia and *what* they write about.

Conditioning on topic alone opens a backdoor path through country and ownership.

► Solution

The Solution: Conditioning on Source and Topic

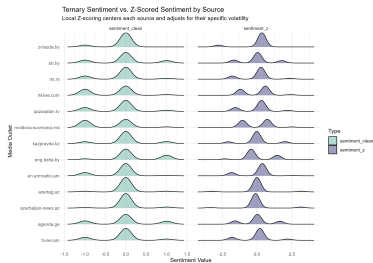


Source identity subsumes country and ownership; source-level z-scoring plus topic fixed effects blocks the backdoor paths. Because topic is post-treatment, the topic-adjusted estimates are *controlled direct effects*, which require sequential unconfoundedness beyond continuity (Acharya, Blackwell & Sen 2016). Total effects without topic adjustment are substantively similar: H2 0.208 vs. 0.222; H3 -0.37 vs. -0.33 . Variants using only pre-invasion topic and source adjustments hold as well.

Source-Level Z-Score Standardization

Problem: source fixed effects absorb the group indicator, so the RDID interaction cannot be estimated; and high-variance sources would dominate a linear specification.

Solution: standardize sentiment within each source over the *full sample period*. Demeaning is equivalent to a source fixed effect (Frisch–Waugh–Lovell) but leaves G_p estimable; dividing by σ_s equalizes source influence and puts coefficients in within-source SDs. For H1, the z-score and full fixed-effects specifications agree.



Inference with Few Clusters

Challenge: source-level clustering yields 60 clusters for H2 and 13 for H3.

- H1: cluster-robust SEs from `rdrobust` (61 clusters)
- H2/H3 interactions: wild cluster restricted bootstrap (Cameron, Gelbach & Miller 2008), Webb six-point weights, $B = 99,999$, null imposed
- Country-level clustering (11 / 8 clusters) leaves H2 and H3 significant ($p = 0.021$, $p = 0.032$)
- Saturated source-by-topic RDDs aggregated by pairs and Bayesian bootstrap sidestep clustering; results consistent

Abadie et al. (2022): if these outlets are the population rather than a sample, clustering is conservative. We report clustered inference regardless.

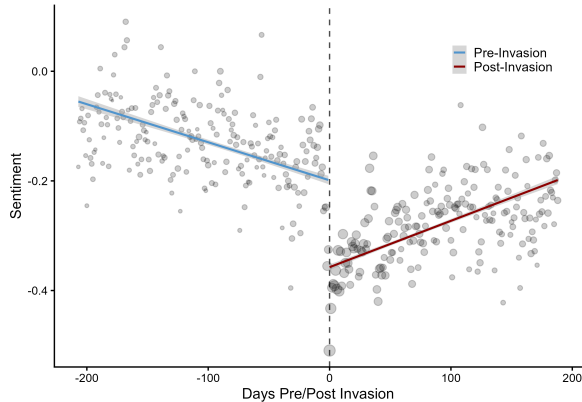
The Forcing Variable

- Distance from publication *date-time* to the invasion *date*; every article on or after February 24 is treated, so the design stays sharp
- Date-time avoids the mass points of calendar dates; missing timestamps are imputed by kernel density over the source's observed publication times, drawn once and held fixed

Alternative constructions:

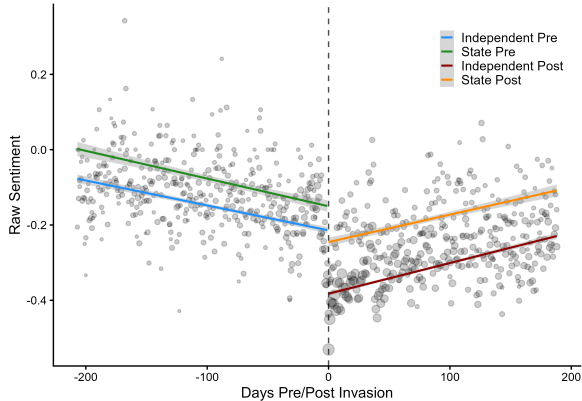
- Date only (mass points present): substantively identical; H3 marginal ($p = 0.06$ – 0.10)
- Drop imputed articles (about a quarter of the sample): signs unchanged; H2 at bootstrap $p = 0.10$

Full-Sample Figures (Military Articles Included)



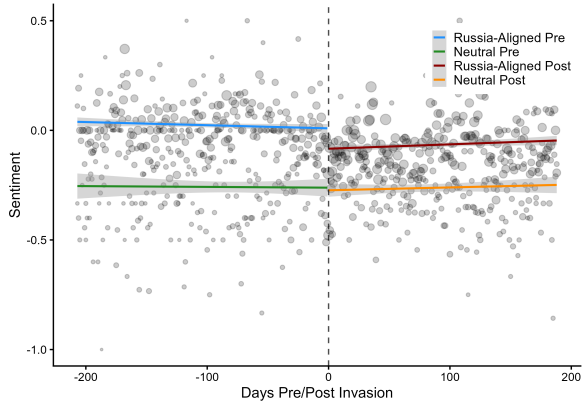
H1, all articles (Table 1, Model 5: -0.217 SD).

Full-Sample Figures (Military Articles Included)



H2, all articles (Table 2, Model 3: interaction 0.080, n.s.).

Full-Sample Figures (Military Articles Included)



H3, all articles (Table 4, Model 3: interaction -0.19 , n.s.).

D. Results Tables

H1: Robust Across Specifications

	(1) Baseline (raw)	(2) Topic + Source FEs	(3) Z-Score	(4) Z-Score Non-Mil.	(5) Z-Score No NATO
Invasion	-0.130*** (0.022)	-0.106*** (0.015)	-0.217*** (0.036)	-0.184*** (0.033)	-0.284*** (0.038)
Clusters	61	61	61	61	45

All models use the MSE-optimal bandwidth; SEs clustered at the source level. (1)–(2) raw sentiment; (3)–(5) source-level z-score with topic FEs. (4) omits military topics; (5) omits military topics and NATO allies. Main-deck estimate is (4).

H2: State Ownership Interaction

	(1) [‡] State	(2) [‡] Indep.	(3) [†] Full	(4) [†] Non-Mil.	(5) [†] Non-NATO	(6) [†] Matched
Invasion	-0.109+ (0.058)	-0.214*** (0.034)	-0.253*** (0.036)	-0.240*** (0.031)	-0.347*** (0.027)	-0.348*** (0.026)
State-Owned × Treat			0.080 (0.063)	0.14+ (0.06)	0.222* (0.069)	0.222* (0.069)
Clusters	17	44	60	60	43	43

[‡]Separate RDDs (non-military), cluster-robust SEs. [†]WCR bootstrap ($B = 99,999$). (3) full sample, optimal bandwidth. (4) omits military topics. (5) also omits NATO allies. (6) matching on topic and date, applied to (5). All z-scored. Main-deck estimate is (6).

H2: State Management Varies by Topic

	Political	Military	Economic	Cultural	Other
Invasion	−0.213*** (0.027)	−0.422*** (0.068)	−0.423*** (0.066)	−0.547*** (0.071)	−0.269*** (0.039)
State-Owned × Treat	0.155 (0.079)	0.012 (0.102)	0.290* (0.130)	0.100 (0.190)	0.270** (0.100)

The interaction is positive in every topic, significant for Economic and Other. The divergence is present *within* topics, so it is not an artifact of compositional change.

H3: Russia-Aligned Interaction

	(1) [‡] Aligned RDD	(2) [‡] Neutral RDD	(3) [†] Full	(4) [†] Non-Mil.	(5) [†] Matched
Invasion	-0.220* (0.099)	0.055 (0.101)	-0.068 (0.080)	0.098 (0.070)	0.098 (0.069)
Russia-Aligned $\times$ Treat			-0.19 (0.11)	-0.33* (0.11)	-0.38* (0.11)
Clusters	9	4	13	13	13

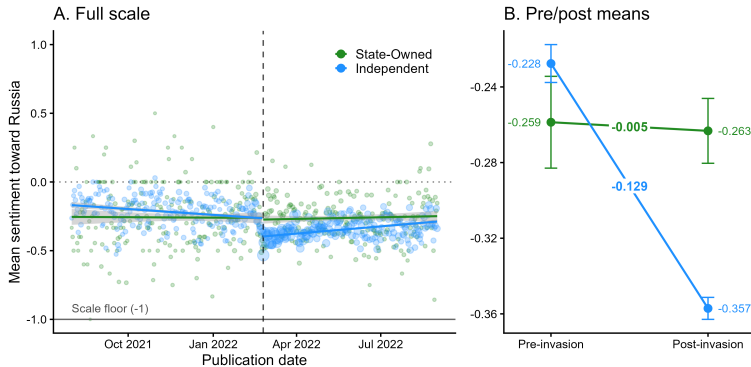
[‡]Separate RDDs (non-military) with cluster-robust SEs. [†]WCR bootstrap ($B = 99,999$). Sample restricted to state-owned media in non-NATO countries. (3) full sample at optimal bandwidth. (4) omits military topics. (5) matching on topic and date, applied to (4). All z-scored. Main-deck estimate is (5).

Alternative Partition: Ethnic Russian Population Share

	(1) [‡] High RDD	(2) [‡] Low RDD	(3) [†] Full	(4) [†] Non-Mil.	(5) [†] Matched
Invasion	-0.079 (0.066)	-0.108 (0.102)	-0.17 (0.10)	-0.13 (0.10)	-0.143 (0.099)
High Russian × Treat			-0.055 (0.133)	-0.002 (0.148)	0.015 (0.145)

Same sample, same method: a significant differential when states are split by security alignment, none when split by ethnic Russian share. Inconsistent with pretext exposure or cultural proximity driving H3. [► Cross-Tab](#)

Neutral-Country Media Relative to the Scale Floor



Neutral state media sits far above the scale floor of -1 and moves by -0.005 across the invasion, while independent outlets in the same countries drop by -0.129 . The absence of a neutral state-media response is not a floor or measurement artifact.

E. Robustness

Placebo Test: December 7, 2021

Russia-aligned state media, non-military, by country and pooled (rdrobust, z-scored)

	Armenia	Azerbaijan	Belarus	Kazakhstan	All Aligned
Estimate	0.095	-0.163	0.320*	-0.172	0.108
Std. Error	(0.223)	(0.143)	(0.143)	(0.265)	(0.089)
p-value	0.669	0.254	0.025	0.518	0.225
Bandwidth (days)	21.8	18.7	20.9	19.5	27.9

Only Belarus — the staging ground — shows a break, and it is *positive*: a rally to the patron at the announcement that faded as the invasion became more plausible. Pre-invasion data only; rdrobust default SEs because each country has 2–3 state-owned sources.

Placebo Test: December 7, 2021

Full-sample placebo across the three designs

Model	Estimate	SE	p
H1: Full model (RDD)	0.016	(0.033)	0.637
H1: Z-score (RDD)	0.000	(0.063)	0.994
H2: Diff-in-Disc	0.250	(0.119)	0.087
H3: Non-military	0.705	(0.287)	0.143

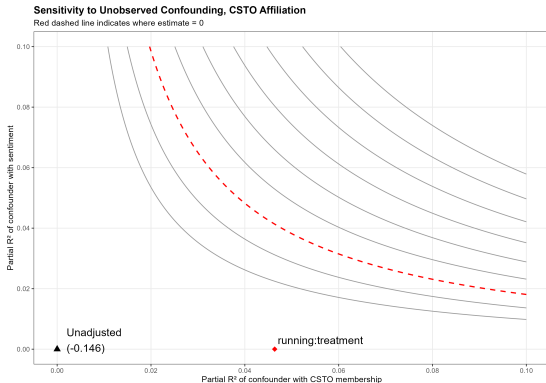
December 7, 2021 (US public warning) as a false treatment date, data restricted to the pre-invasion period. No significant discontinuity in any design. December 7 also lies about three bandwidths (19–28 days) from the true cutoff, so any response to the warning cannot mechanically contaminate the February 24 estimates.

H3: Leave-One-Out Country Robustness

<i>Standard Model</i>				<i>Matched Model</i>			
Dropped	Coeff.	SE	p	Dropped	Coeff.	SE	p
Belarus	−0.188	(0.136)	0.201	Belarus	−0.203	(0.142)	0.186
Armenia	−0.258*	(0.102)	0.030	Armenia	−0.344*	(0.109)	0.010
Georgia	−0.230*	(0.101)	0.043	Georgia	−0.280*	(0.107)	0.024
Azerbaijan	−0.361*	(0.136)	0.024	Azerbaijan	−0.387*	(0.136)	0.017
Kazakhstan	−0.249*	(0.085)	0.015	Kazakhstan	−0.277**	(0.076)	0.004
Moldova	−0.326*	(0.114)	0.015	Moldova	−0.383**	(0.110)	0.005
Kosovo	−0.298	(0.138)	0.054	Kosovo	−0.348*	(0.130)	0.021
Serbia	−0.387**	(0.112)	0.005	Serbia	−0.407**	(0.112)	0.004

Bandwidth recalculated for each subset; SEs clustered at the source level. Belarus attenuation is expected — as the most constrained state, its removal reduces the signal. The result is otherwise stable across all exclusions; source-level leave-one-out is likewise stable.

Sensitivity to Unobserved Confounding

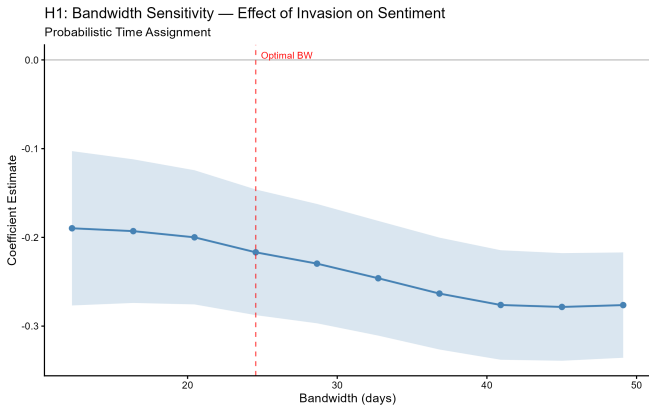


Matched sensitivity procedure: coarsened exact matching on date and topic, then Cinelli & Hazlett (2020) on the RDID interaction, with bootstrapped SEs.

Benchmark: even granted a 56% association with the interaction, a confounder would still need $1.5\times$ the invasion's own association with sentiment to drive the differential to zero.

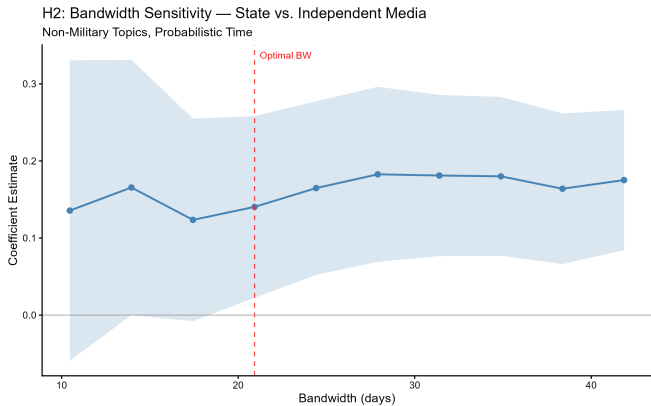
A qualifying confounder must vary differentially at the cutoff by group, since standardization absorbs time-invariant traits. Robustness value 1.9% (small because topic FEs leave little residual variance).

Bandwidth Sensitivity



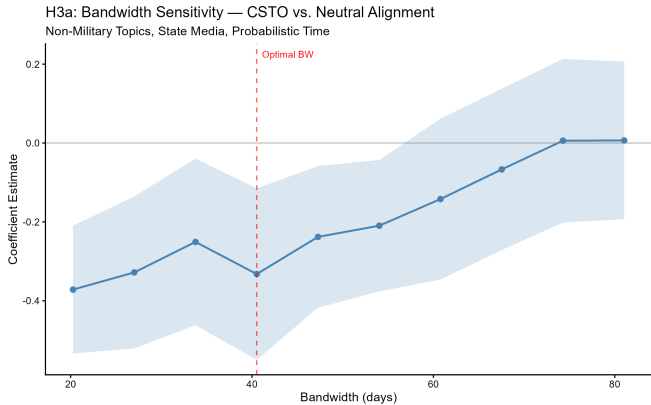
H1. Red dashed line marks the MSE-optimal bandwidth; shaded region is the 95% confidence interval. Results are stable from $0.5\times$ to $1.5\times$ the optimal bandwidth.

Bandwidth Sensitivity



H2 interaction. Red dashed line marks the MSE-optimal bandwidth; shaded region is the 95% confidence interval. Results are stable from $0.5\times$ to $1.5\times$ the optimal bandwidth.

Bandwidth Sensitivity



H3 interaction. Red dashed line marks the MSE-optimal bandwidth; shaded region is the 95% confidence interval. Results are stable from $0.5\times$ to $1.5\times$ the optimal bandwidth.

F. Discussion

Western Trade Shares: Russia-Aligned States

Country	2021	2022
Azerbaijan	72%	73%
Kazakhstan	46%	48%
Armenia	34%	29%
Belarus	22%	(disrupted by sanctions)

Western trade shares (Europe + North America as % of total trade) remained broadly stable through 2022. Consistent with the interpretation that these states were not categorized alongside Russia by Western governments — though we cannot establish a direct causal link to the media signal. Source: World Bank (2021, 2022).